Aayusha Bhatta

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PROFESSIONAL SUMMARY

Motivated Computer Engineering graduate (IOE - KEC) with hands-on experience in web development, machine learning, and cybersecurity. Research-published at a national conference, with demonstrated experience in penetration testing, automated testing frameworks, and data visualization. Consistently improving academic trajectory (top semester at 90%). A self-driven, research-oriented professional eager to contribute technical skills and grow within a dynamic software development environment.

RESEARCH EXPERIENCE

Research Collaborator (Remote) — Aug 2026 - Present

Wayne State University | Advisor: ***Prof. Lihui Liu***

- Conducted a comprehensive literature survey on Intelligent Recommendation Systems in Education.
- Drafted a 50-paper survey with 9 sections, applying PRISMA methodology for systematic review and citation management.
- Collaborating on manuscript preparation for upcoming publication.

OMAMA-DB Breast Cancer Detection — [GitHub Repo]

UMass Boston (Machine Psychology Lab) | Independent Research Project

- Built a study-level multiview classification pipeline using the OMAMA-DB dataset.
- Refactored single-view baseline to group 4 views (CC/MLO) per study, stacking them into a 4-channel tensor (Early Fusion).
- Applied ROI cropping using JSON metadata to eliminate background artifacts, jumping validation accuracy from 66.60% to 99.40%.

ACADEMIC PROJECTS

MediPredict - Multiple Disease Prediction System — [GitHub Repo]

Final Year Major Project | Role: Flask Backend & Frontend Developer

- Led the Flask backend and frontend development for an ML-powered web app predicting diabetes, heart disease, and Parkinson's disease.
- Integrated classification algorithms achieving high prediction accuracy; collaborated on model selection and data pipeline design.
- Research paper presented at KHWOPA CEEL 2026 National Conference, Bhaktapur, Nepal (February 2026).

Churn It - Customer Retention Prediction Model — 2024

3rd Year Minor Project | Role: Frontend Developer

- Developed the frontend interface (HTML/CSS) for a customer churn prediction web app built for a Nepali company.
- Collaborated with team members handling ML models (Random Forest, XGBoost) to integrate backend predictions into the frontend display.

EDUCATION

B.Tech - Computer Science & Engineering 2022 - 2026

Kathmandu Engineering College (KEC) / IOE, TU | Overall Aggregate: 78.78%

- **Semester Performance:** I: 62.48%, II: 79.08%, III: 69.94%, IV: 78.89%, V: 77.60%, VI: 83.54%, VII: 80.48%, VIII: 90.00% (Topper)
- +2 Science GPA: 3.4/4.0 | SEE GPA: 3.8/4.0

TECHNICAL SKILLS

Languages & Web: Python, JavaScript, C, HTML, CSS, SQL

AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Random Forest, XGBoost, NLP Fundamentals

Frameworks & Tools: Flask, React (learning), Git, Linux, Power BI, Excel

Security: Penetration Testing, Vulnerability Assessment, Security Test Automation